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Roll no -69

Experiment 4

Title

Develop adynamic webpage where content updates on button click using JavaScript DOM. ImplementjQuery effects: hide/show elements, toggle visibility, and animate elements.

Theory:

1. **JavaScript and the DOM (Document Object Model):**
   - The **DOM** is a hierarchical representation of theelements on a webpage. It allows programming languages like JavaScript to interactwith the structure, style, and content of the page.
   - Using **DOM methods**, we can:
     - Access elements (getElementById, getElementsByClassName, querySelector)
     - Change content (innerHTML, textContent)
     - Modify attributes (setAttribute, removeAttribute)
     - Change styles dynamically (style.property)
   - Example: Changing the text of a paragraph when a button is clicked enhances user interactivity.
2. **Dynamic Webpage Content:**
   - A **dynamic webpage** changes content without reloading the page.
   - Dynamic updates improve **user experience** by allowing real-time feedback and interactivity.
   - In this lab, content updates are achieved using document.getElementById().innerHTML to manipulate text dynamically.
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3. **Introduction to jQuery:**

- **jQuery** is a lightweight, fast JavaScript library that simplifies:

HTML document traversal and manipulation

  - Event handling
  - Animations and effects

- jQuery is **cross-browser compatible**, meaning the same code works in different browsers without modification.

Syntax Example:

$(selector).action();

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  - $() is the jQuery function
  - selector identifies theHTML element
  - action() is the methodapplied (like hide(), show(), animate())

4. **jQuery Effects and Animations:**
   - **Hide/Show:** Instantly hideor show elements.
   - **Toggle:** Switch visibility state of an element (hide → show, show → hide).
   - **Animate:** Allows smooth transitions of CSS properties (like height, width, opacity, margins).
   - **Benefits of jQuery Effects:**
     - Improves visual interactivity
     - Reduces manual CSS manipulation
     - Provides smoother user experience
5. **Integration of JS DOM and jQuery:**

- Using **pure JavaScript** allows control over element content and structur

- Using **jQuery**complements JS by handling animations and effects more effici ently.

- Combiningbothallows for **dynamic and interactive web pages** where content changesanduserinteractions are smooth and visually appealing.

6. **Applications:**

   - Dynamicwebpages are used in:

     - Onlineforms that validate inputs instantly
     - Contentsliders and carousels
     - Real-timenotifications
     - Interactivedashboards

Procedure

1. Include jQuery CDN:

```
<script src="https://code.jquery.com/jquery-3.7.0.min.js"></script>
```

2. Create HTML element:

```
<div id="box">Hello</div>
<button id="btn">Toggle</button>
```

3. Apply jQuery:

```
<script>
$('#btn').click(function(){
  $('#box').toggle(500);
});
</script>
```

Output

Dynamic Webpage using JavaScript DOM and jQuery
Click the button to change this text.
Change Text
Hello
Hide
Show
Toggle
Fade
Animate
Stop

Dynamic Webpage using JavaScript DOM and jQuery
The content has been updated dynamically!
Change Text
Hello
Hide
Show
Toggle
Fade
Animate
Stop

Dynamic Webpage using JavaScript DOM and jQuery
The content has been updated dynamically!
Change Text
Hide
Show
Toggle
Fade
Animate
Stop

Conclusion

JavaScript DOM allows dynamic updating of webpage content in real-time. jQuery simplifies element manipulation, effects, and animations. Combining both creates interactive and visually appealing webpages, enhancing user experience. This lab demonstrated practical implementation ofcontent updates, visibility toggling, and animations.

Question:

1. Whatis the difference between .hide() and .toggle() in jQuery?
2. Howcan you **stop an ongoing animation** in jQuery before it completes?
3. Howcan you use jQuery to **fade in** and **fade out** an element?